

Colour, image and identity

Building a restrained visual system, and preparing images with useful alternative text and appropriate crops.



Colour, image and identity

Build a token system

Define 5-7 semantic colour tokens in HSL, not scattered hex codes, and check every text/background pairing against WCAG contrast ratios.

Prepare real images

Choose crops by role (hero, portrait, avatar, decorative) and reason about aspect-ratio, object-fit and Cumulative Layout Shift.

Write useful alt text

Classify an image as decorative, informative, functional or complex before writing its alternative text.

From taste to a system

Two photographs, one idea



Why tokens, not scattered hex

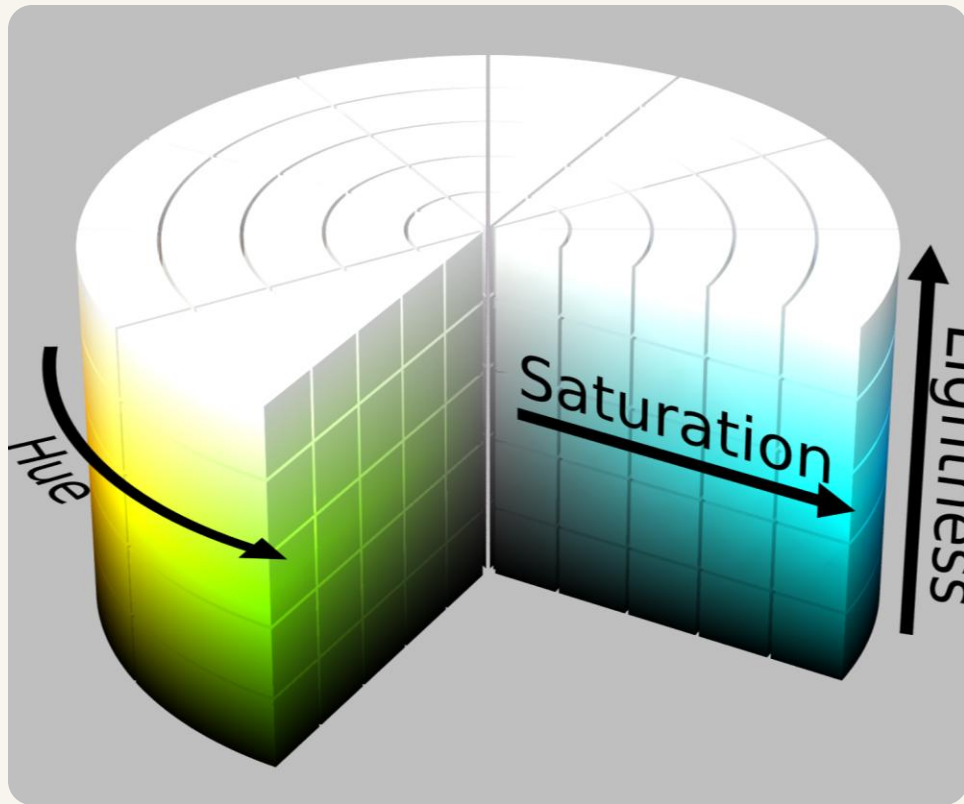
Name the role, not the colour

A hex code like #C7831A tells CSS what to paint. It says nothing about why. A design token adds a name for the role that colour plays: --color-accent, --color-text, --color-surface.

```
/* scattered */
h1 { color: #18212A; }
.btn { background: #C7831A; }
.card { border: 1px solid #D7D0C4; }

/* tokenised */
:root {
  --color-text: #18212A;
  --color-accent: #C7831A;
  --color-border: #D7D0C4;
}
h1 { color: var(--color-text); }
.btn { background: var(--color-accent); }
.card { border: 1px solid var(--color-border); }
```


The same colour, two notations



`hsl(36, 77%, 44%)` is the SLOP1512 gold token. Hex hides three separate decisions inside six characters; HSL exposes them.

Hue (36°)

Position on the colour wheel. 36° sits between orange and amber.

Saturation (77%)

Colourfulness. High saturation reads as vivid rather than muted.

Lightness (44%)

Distance from black to white. 44% is a mid-dark tone.

Why it matters

Need a lighter accent for dark mode? Raise lightness and keep hue and saturation fixed.

The SLOP1512 token set

Token	Hex	HSL	Role
--color-ink	#18212A	hsl(210, 27%, 13%)	Primary text, dark surfaces
--color-cream	#F8F5EF	hsl(40, 39%, 95%)	Page background (light)
--color-gold	#C7831A	hsl(36, 77%, 44%)	Accent — never as body text on cream
--color-slate	#43515D	hsl(208, 16%, 31%)	Secondary text, captions
--color-line	#D7D0C4	hsl(38, 19%, 81%)	Borders and rules only
--color-white	#FFFFFF	hsl(0, 0%, 100%)	Cards on dark surfaces

The maths behind 'readable'

Relative luminance (per sRGB channel c in $[0,1]$):

```
c_lin = c / 12.92          if c <= 0.03928
c_lin = ((c + 0.055) / 1.055) ^ 2.4    otherwise

L = 0.2126*R_lin + 0.7152*G_lin + 0.0722*B_lin
```

Contrast ratio between the lighter and darker of two colours:

```
contrast = (L_lighter + 0.05) / (L_darker + 0.05)
```

4.5 : 1

Minimum for normal body text (WCAG 2.1, SC 1.4.3).

3 : 1

Minimum for large text (18pt, or 14pt bold) and for UI components / graphical objects (SC 1.4.11).

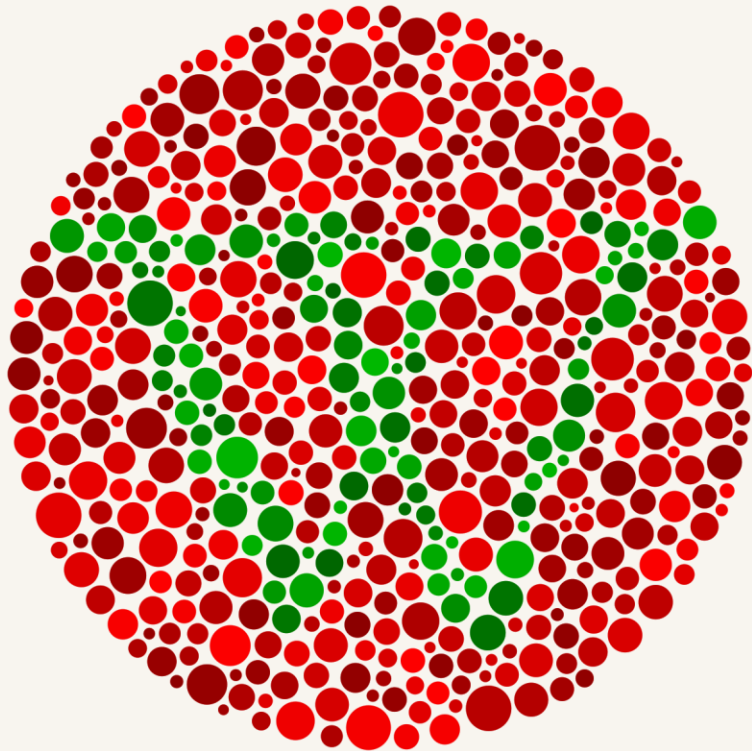
Contrast, worked

Which token pairs actually pass?

Pairing	Ratio	4.5:1 text	3:1 large/UI
ink text / cream bg	14.96:1	PASS	PASS
slate text / cream bg	7.50:1	PASS	PASS
gold text / ink bg	5.19:1	PASS	PASS
gold text / cream bg	2.88:1	FAIL	FAIL
gold icon/border / white bg	3.14:1	FAIL	PASS
slate text / ink bg	2.00:1	FAIL	FAIL

Finding: gold must never carry body text on cream (2.88:1). It is safe as text on ink (5.19:1), and safe as a large graphic/UI accent on white (3.14:1) — but not as white-background text.

Contrast is not colour vision



Protanopia / deuteranopia

Roughly 1 in 12 men and 1 in 200 women have reduced sensitivity to red or green. Red/green status pairs (error/success) can become indistinguishable.

The trap

A high WCAG contrast ratio only measures lightness difference. Red #C0392B and green #27AE60 can both pass 4.5:1 against white and still look identical to a deuteranope.

The fix

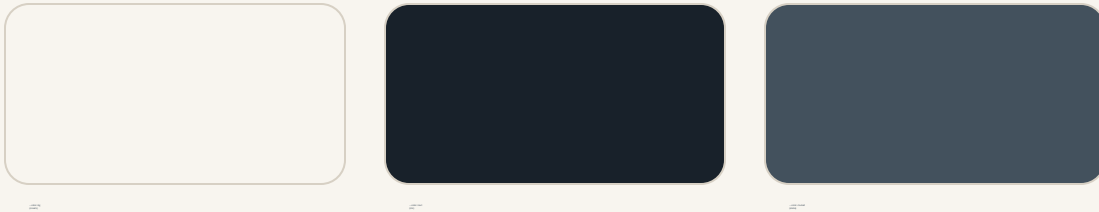
Pair colour with a second channel: an icon, a pattern, an underline, or text ('Error:' / 'Saved') — never colour alone.

Case study

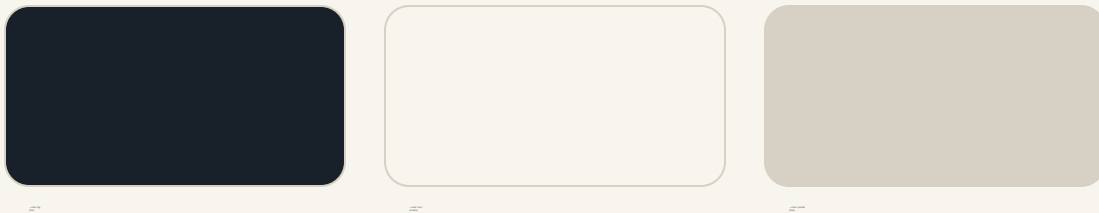
Dark-mode tokens: same roles, re-checked

Dark mode is not 'invert everything' — it is re-pointing the same semantic roles at a palette that still passes contrast.

Light mode



Dark mode



Re-checked ratios

Light: text/bg = ink/cream = 14.96:1.

Dark: text/bg = cream/ink = 14.96:1 (same pair, reversed — ratio is symmetric).

Dark muted: line/ink = 10.63:1 — safe for secondary text on dark, unlike slate/ink at 2.00:1.

Before / after

An unrestrained board vs a token system



Before: any of dozens of hues could appear anywhere, with no way to predict which pairing will occur next. After: exactly the tokens on the table three slides back, and nothing else — every colour decision traces back to one of six names.

Decorative, informative, functional, complex



Decorative

The marble image above carries no information a reader needs — `alt=""` and it is skipped by assistive technology.

Informative

The HSL cylinder diagram earlier this deck communicates real content — its alt text described hue, saturation and lightness.

Functional images (icon-only buttons) describe the action they trigger. Complex images (charts, diagrams with detail) need a short alt plus a longer text description nearby.

object-fit and aspect-ratio

```
.card-media {  
  aspect-ratio: 16 / 9;  
  width: 100%;  
  object-fit: cover;  
  object-position: 50% 30%;  
}
```

aspect-ratio

Reserves the box's shape before the image loads. Prevents the surrounding layout from resizing when the file arrives.

object-fit: cover

Fills the box, cropping overflow. object-position chooses which part of the image survives the crop.

Portrait crop: keep the subject, lose the noise



The rule

Crop so the eyes sit on the upper third line, not the exact centre. A face-safe crop keeps eyes, not the chin or hairline, as the anchor point when the box is later resized.

```
img.avatar-wide {  
  aspect-ratio: 4 / 5;  
  object-fit: cover;  
  object-position: 50% 20%;  
}
```


Square avatar: a tighter ratio, same logic



Why square is harder

A 1:1 crop leaves less room than 4:5 — the same face-safe rule now has to compete with cropping in from both sides, not just top and bottom.

```
img.avatar-square {  
  aspect-ratio: 1 / 1;  
  object-fit: cover;  
  object-position: 50% 25%;  
}
```

Same object-position idea, moved slightly further up to compensate for the narrower box.

srcset and sizes

```

```

How the browser decides

srcset lists candidate files with their real pixel width (480w). sizes tells the browser how wide the image will actually be laid out at the current viewport, so it can pick the smallest file that is still sharp enough — before it knows the final layout.

picture/source: art direction and modern formats

```
<picture>
  <source type="image/avif" srcset="hero.avif">
  <source type="image/webp" srcset="hero.webp">
  
</picture>
```

Format fallback

The browser picks the first `<source>` it can decode. AVIF and WebP are typically 25-50% smaller than JPEG at equal visual quality; the `<img>` is the guaranteed fallback.

Art direction

`picture/source` can also swap to a different crop (not just a format) at a breakpoint — a portrait crop on mobile, a wide crop on desktop, from the same `<img>` alt text.

Reserve the space before the image loads

Layout shift score = impact fraction × distance fraction

Example: an image box covers 60% of the viewport.
It loads late and pushes content down by 180px in a 900px viewport.

```
impact fraction  = 0.60  
distance fraction = 180 / 900 = 0.20  
layout shift score = 0.60 * 0.20 = 0.12
```

The fix

Give the `<img>` explicit width and height attributes (or a CSS aspect-ratio). The browser then reserves 180px of space immediately, so impact fraction on load becomes 0 and no shift occurs — a single score of 0.12 already exceeds the 0.1 'good' CLS threshold, so this is not a minor issue.

Alt text

A decision tree before you write anything

Role	Alt text rule	Example
Decorative	alt="" (empty, but present)	Background texture, marble photo
Informative	Describe what a sighted reader learns	"HSL colour model shown as a cylinder..."
Functional	Describe the action, not the picture	"Search" for a magnifying-glass icon button
Complex	Short alt + adjacent long description	A chart: alt names the trend, caption gives the data

Alt text, worked

Four of this deck's own images

Image	Role	Alt text used in this deck
Chefchaouen street	Informative	"Narrow street in Chefchaouen, Morocco, its walls and stairs painted in closely related shades of blue."
Ishihara plate	Informative	"Ishihara colour-vision test plate: a numeral formed from dots that share lightness but differ in hue..."
HSL cylinder diagram	Informative	"Diagram of the HSL colour model drawn as a cylinder: hue around the rim, saturation from centre outward, lightness bottom to top."
Marble close-up	Decorative	alt="" — the image is used only as visual texture on the image-roles slide

Alt text for identity imagery

Hero and brand images are still informative



The temptation

Hero banners feel decorative, so it is tempting to leave alt="". But if the image sets the tone for the page (a course, a brand, a place), a screen-reader user loses that context entirely.

Write alt text that names the scene, the way a caption would — not a mood ('breathtaking view'), and not a repeat of the heading text next to it.

Logo and brand-mark treatment rules

Clear space

Reserve a margin around the mark at least equal to the height of its own smallest element. Never let text or other graphics enter that margin.

Minimum size

Below a stated minimum width, fine details in a mark disappear or merge. State the minimum in the brand guide, not by trial and error each time.

Do not recolour

A mark's colours are part of its identity. Recolouring it to 'match the page' undermines the one thing a logo is supposed to do: stay recognisable.

Contrast against backgrounds

Provide light and dark variants of the mark rather than placing one variant on a background it was not designed for — apply the same 3:1 UI-component threshold from earlier.

Studio task

Build your own restrained system

Part 1 — tokens

Define 5-7 semantic colour tokens (not raw hex scattered through your CSS). Check every text/background pairing you actually use against the correct WCAG threshold (4.5:1 body text, 3:1 large text/UI). Provide both a light and a dark variant.

Part 2 — images

Select two real images for your own page. For each: state its role (decorative/informative/functional/complex), choose a crop and aspect-ratio with a stated reason, and write alt text that matches its role.

Part 3 — evidence

Record: the token table with hex + HSL + contrast ratio; the source, licence and access date for each image; and the final alt text you chose, with one sentence justifying it.

Studio task

Rubric

Criterion	What passes
Token discipline	5-7 named tokens used consistently; no stray hex values in the final CSS
Contrast evidence	Every real text/background pairing checked against the correct threshold, with the ratio shown
Crop justification	Aspect-ratio and object-position chosen for a stated reason, not left at defaults
Alt text fit	Alt text matches the declared role (decorative/informative/functional/complex)
Source record	Licence, author and access date recorded for every real image used

Source record

Images used in this deck (1 of 2)

File / local name	Wikimedia Commons source	Licence · author
w5-chefchaouen.jpg	Chefchaouen, Morocco (Unsplash).jpg https://commons.wikimedia.org/wiki/File:Chefchaouen,_Morocco_(Unsplash).jpg	CC0 1.0 Universal · Azer Koçulu
w5-burano.jpg	Burano Houses (8492557865).jpg https://commons.wikimedia.org/wiki/File:Burano_Houses_(8492557865).jpg	CC BY 2.0 · Paul VanDerWerf
w5-hsl-cylinder.png	HSL color solid cylinder saturation gray.png https://commons.wikimedia.org/wiki/File:HSL_color_solid_cylinder_saturation_gray.png	CC BY-SA 3.0 · SharkD
w5-ishihara.png	Ishihara-Test.svg https://commons.wikimedia.org/wiki/File:Ishihara-Test.svg	CC BY-SA 3.0 · Dan-yell
w5-face-portrait.jpg	Face portrait (Unsplash).jpg https://commons.wikimedia.org/wiki/File:Face_portrait_(Unsplash).jpg	CC0 1.0 Universal · William Stitt
w5-headshot.jpg	Headshot.jpg https://commons.wikimedia.org/wiki/File:Headshot.jpg	CC0 1.0 Universal · John A. Hollansworth Jr.

Source record

Images used in this deck (2 of 2)

File / local name	Wikimedia Commons source	Licence · author
w5-rainbow-swatches.jpg	RAL Color Card (2354626166).jpg https://commons.wikimedia.org/wiki/File:RAL_Color_Card_(2354626166).jpg	CC BY-SA 2.0 · CoCreatr
w5-farrowball-grey.jpg	Minnesota ten color palat september wiki version.jpeg https://commons.wikimedia.org/wiki/File:Minnesota_ten_color_palat_september_wiki_version.jpeg	CC BY-SA 3.0 · Firry Floyd
w5-marble.jpg	A Close-Up of an Intricate Marble Pillar.jpg https://commons.wikimedia.org/wiki/File:A_Close-Up_of_an_Intricate_Marble_Pillar.jpg	CC BY-SA 4.0 · Pim58
w5-andes-panorama.jpg	Andes mountains panoramic view.jpg https://commons.wikimedia.org/wiki/File:Andes_mountains_panoramic_view.jpg	CC BY-SA 3.0 / GFDL · Paolo Costa Baldi
w5-workspace.jpg	Laptop and coffee on a desk.jpg https://commons.wikimedia.org/wiki/File:Laptop_and_coffee_on_a_desk.jpg	CC0 1.0 Universal · Lauren Mancke

All images accessed 13 September 2026. 11 distinct images in total, all placed as slide images in this deck.

Review

Checklist and next week

Before you ship

Every colour in your CSS traces to a named token.
Every text/background pairing you use passes 4.5:1 (or 3:1 for large text/UI).
No status or meaning relies on colour alone.
Every image has a stated role and alt text that matches it.
Every real photo has a recorded source, licence and access date.

Next week

Typography and layout rhythm: how type scale, spacing and grid extend the same token discipline from colour into every other visual decision on the page.